

Problems Proposed for 2025-26 Pal4Pal

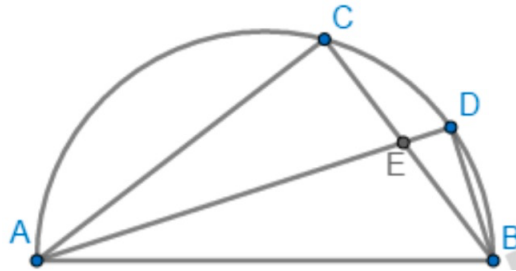
Steven, Derek, Andy

August 2025

Steven

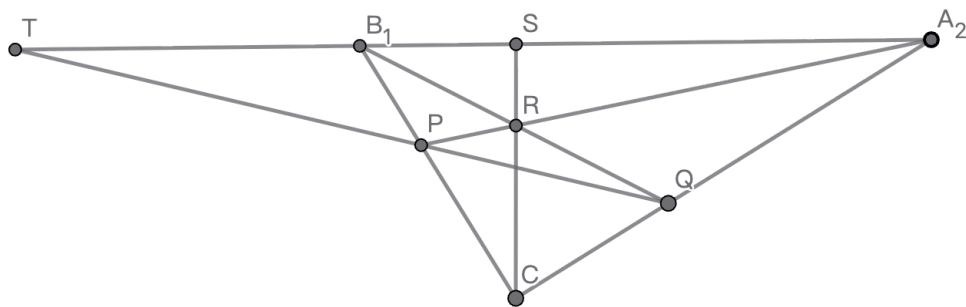
Q1 Geometry [4 points]

Consider semicircle with diameter AB , C is a point, D is the midpoint of arc BC . AD meets BC at E . If $CE = 3$, and $BD = 2\sqrt{5}$, find AB .



Q2 Geometry [4 points]

Consider right $\triangle ABC$ with $\angle C = 90^\circ$, $AB = 10$, and $AC = 8$. $CP = CQ = 2$. BQ and AP meet at R . CR meet AB at S . QP meet AB extension at T . Find TS .



Derek

Q1 Combinatorics [4 points]

Ryan has three six-sided dice, one fair, one that rolls threes $\frac{6}{7}$ of the time,

and each of the other five sides equally with probability $\frac{1}{35}$ of the time, and one that rolls threes $\frac{2}{7}$ of the time, and each of the other five sides equally with probability $\frac{1}{7}$ of the time. He selects one of the dice at random. The probability he rolls a three given the first two rolls are threes is $\frac{p}{q}$. What is the remainder when $p + q$ is divided by 1000?

Q2 Number Theory [4 points]

For each integer n , $1 < n < 500$, there exists a number that is divisible by all positive integers less than or equal to n except m and $m + 1$. Find the sum of all possible m .

Andy

Q1 Number Theory [4 points]

Let a and b be positive integers satisfying

$$\text{lcm}(a, b) - \text{gcd}(a, b) = 23$$

Find the sum of all possible values of $a + b$

Q2 Number Theory [4 points]

Let n be a positive integer. Find all sum of all possible values of $\text{gcd}(5n + 11, n + 7)$